
Problem 8.29. Show that A_{NFA} is NL-complete.

Proof. To show that A_{NFA} is NL-complete, we must show that it is in NL and that all NL problems are log space reducible to it. To show $A_{NFA} \in$ NL, we give an algorithm M that decides A_{NFA} in non-deterministic log space.

$M =$ "On input $\langle N, w \rangle$, where $N = (Q, \Sigma, \delta, q_0, F)$ is an NFA and w is a string:

1. If some $w_i \notin \Sigma$, then *reject*.
2. Let q, i and n be binary integers on the work tape, where q is index of q_0, $n = |w|$ and $i = 0$.
3. If $i > n$, then:
4. If q is an accept state in N, then *accept*. Otherwise, *reject*.
5. Non-deterministically, repeat for binary index q_i of each state in N:
6. If $q_i \notin \delta(q, w_i)$, then *reject*.
7. Let $q = q_i$, $i = i + 1$, and go to step 2.
8. *Reject*."

To prove that A_{NFA} is NL-hard, we give a log space reduction from $PATH$ to A_{NFA}. This reduction converts a directed graph G and two nodes s and t to an NFA N and a string w so that G has a directed path from s to t, iff N accepts w. NFA $N = (Q, \Sigma, \delta, q_0, F)$ and w are constructed as follows:

1. Every node in G is a state in N.
2. $\Sigma = \{1\}$.
3. Every edge (u, v) in G is a transition from state u to v by reading symbol 1. Each state in N also contains a transition to it self by reading symbol 1.
4. $q_0 = s$.
5. $F = \{t\}$.
6. Finally, $w = 1^{n-1}$, where n is the number of states in N.

Suppose graph G contains a path $P = (v_1, v_2, \ldots, v_k)$ where $v_1 = s$ and $v_k = t$. Then, each edge in path P is a transition in N. The length of string w equals the maximum number of edges in a longest path in G. Therefore, after reading the last symbol of w, at least one branch of N would be at accept state t.

Suppose NFA N accepts string w. Then, an accepting branch of N on w gives a path in G from s to t.

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